DIVYANSHU YADAV

Computer Science AI & ML

+91-9671498171 | divyanshuyadav715@gmail.com | linkedin.com/in/divyanshu-yadav-2911a52b2 | github.com/divyanshuyadav715-ship-it

EDUCATION DETAILS

Course Name	Institute Name	University Name	Year	Score	Type
B.Tech in Computer Science	Vellore Institute of Technology, Bhopal	VIT	2027	**8.6 CGPA**	Full-time
Class XII	RPS Public School, Rewari	CBSE	2022	92%	Full-time
Class X	RPS Senior Secondary School, Mahendargarh	CBSE	2020	87%	Full-time

PROFESSIONAL SUMMARY

Focused software developer with a strong foundation in data structures and algorithms, proficient in **C++** and **Python**. Experienced in building high-throughput asynchronous inference engines, predictive machine learning pipelines, and actionable data analytics workflows. Equipped with a solid understanding of AWS architecture, database management, and distributed systems.

TECHNICAL SKILLS

Languages: C++, Python, SQL, HTML, CSS
Frameworks & Libraries: FastAPI, PyTorch, Scikit-Learn, XGBoost, Pandas, NumPy, Streamlit
Databases & Caching: Redis (In-Memory / Pub-Sub), SQLite, MySQL / PostgreSQL
Developer Tools: Git, GitHub, VS Code, Locust (Load Testing), Postman
Core Competencies: Data Structures & Algorithms, Object-Oriented Programming (OOP), Asynchronous API Design, High-Throughput Model Serving, Business Metric Optimization

STRONG SUBJECTS

Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Database Management Systems (DBMS), Operating Systems, Cloud Computing

CERTIFICATIONS

- **AWS Training & Certification:** AWS Certificate of Completion (Cloud Architecture & Fundamentals)
- **Coursera:** Applied Machine Learning in Python (University of Michigan)
- **NPTEL:** Cloud Computing (Govt. of India Certified)

PROJECTS UNDERTAKEN

• **InferX – Scalable AI Serving Engine** | *Python, FastAPI, Redis, PyTorch, Locust* 2024

- Engineered an asynchronous deep learning inference gateway separating client API from GPU workers using an event-driven Redis architecture to decouple traffic spikes.
- Implemented **Dynamic Batching** at the worker level with a 10ms sliding window, grouping incoming HTTP requests into unified PyTorch tensor blocks for parallel GPU execution.
- Load-tested with **Locust** to handle 1,000 concurrent simulated users, achieving **309 RPS** with a **p95 latency of 185ms** and 0.00% request failure rate.

• **Predictive Customer Churn & Intervention Engine** | *Python, XGBoost, Scikit-Learn, SQLite, Streamlit* 2024

- Engineered an end-to-end ML classification pipeline analyzing **50,000+ subscriber behavioral logs** to identify at-risk cohorts before cancellation.
- Mitigated severe class imbalance using SMOTE and feature engineering, tuning **XGBoost** to maximize Recall and PR-AUC on churning subscribers.
- Built an automated decision engine comparing **Customer Acquisition Cost (CAC)** against **Lifetime Value (LTV)** to dynamically target retention discounts.

• **E-Commerce Conversion Funnel & Drop-Off Analysis** | *Advanced SQL, Python, Pandas, Tableau / BI Dashboards* 2024

- Simulated and analyzed **100,000+ web traffic logs** using SQL Common Table Expressions (CTEs) and Window Functions (`LEAD`, `LAG`, `ROW_NUMBER`) to pinpoint funnel bottlenecks.
- Built an interactive dashboard to visualize stage-to-stage conversion rates across the user journey (Homepage → Product Page → Cart → Payment).
- Identified a critical 38% mobile checkout UI drop-off, recommending a streamlined UX redesign projected to recapture significant monthly revenue.

CO-CURRICULAR ACTIVITIES & ACHIEVEMENTS

- **Problem Solving:** Solved **200+ problems on LeetCode** utilizing **C++**, strengthening analytical thinking and data structure optimization skills.
- **Leadership:** Managed and coordinated **50+ technical events and workshops** across the university AI club by collaborating with cross-functional student teams.
- **Coordination:** Led planning, communication, and logistics with faculty members, student volunteers, and external participants for seamless event execution.